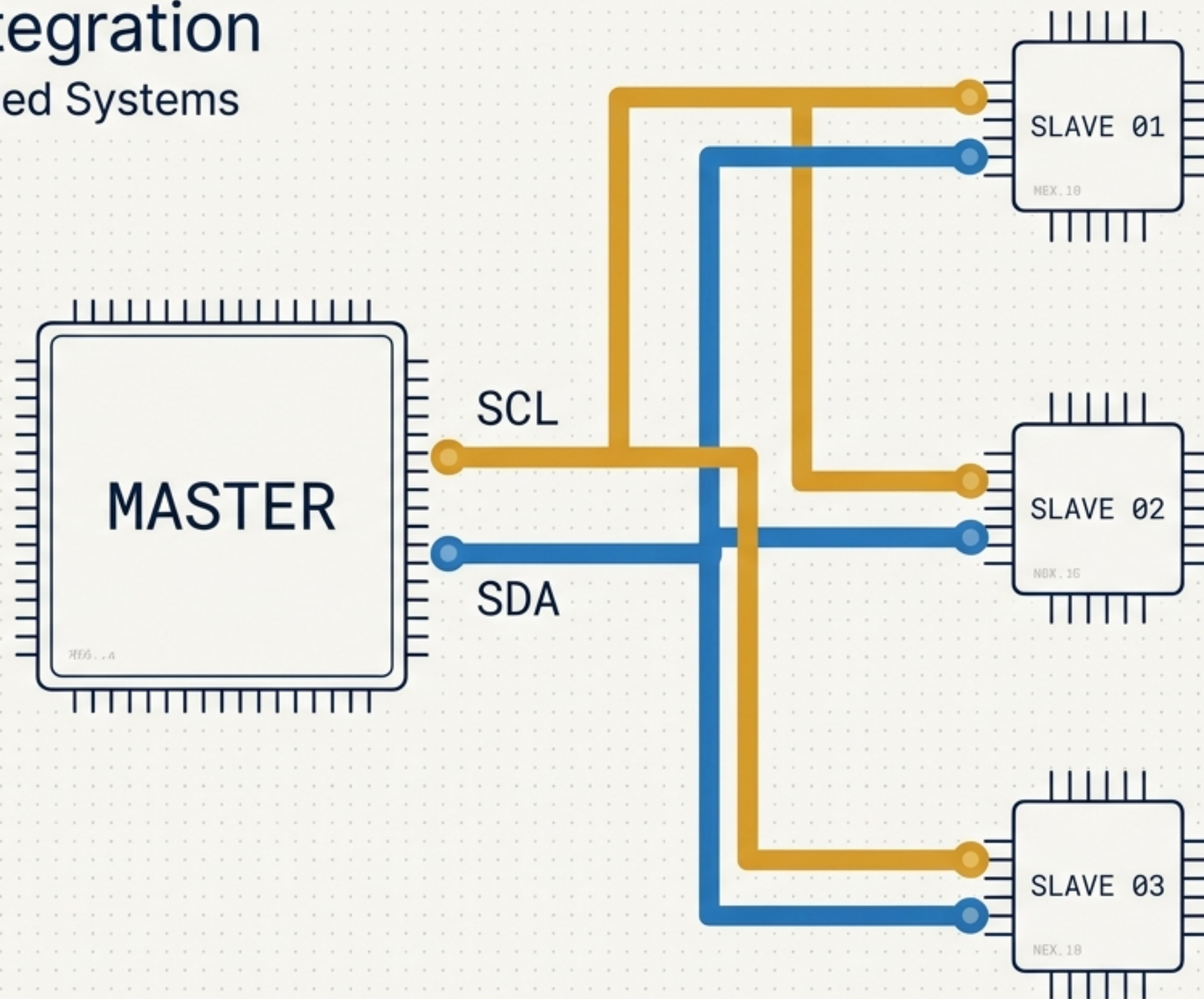


I2C Protocol & Sensor Integration

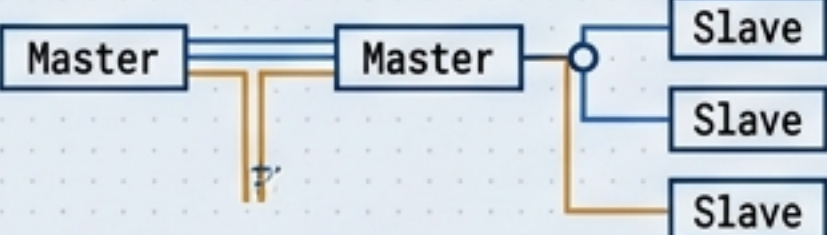
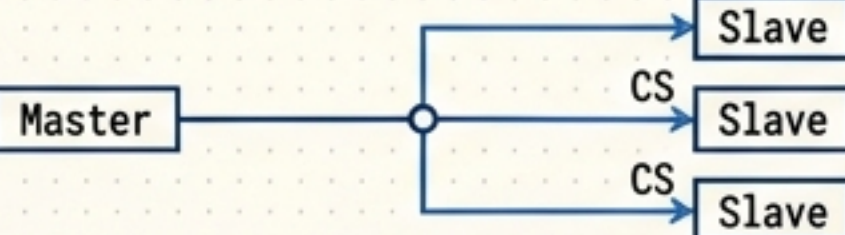
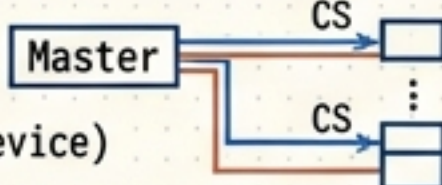
Building Robust Multi-Device Embedded Systems



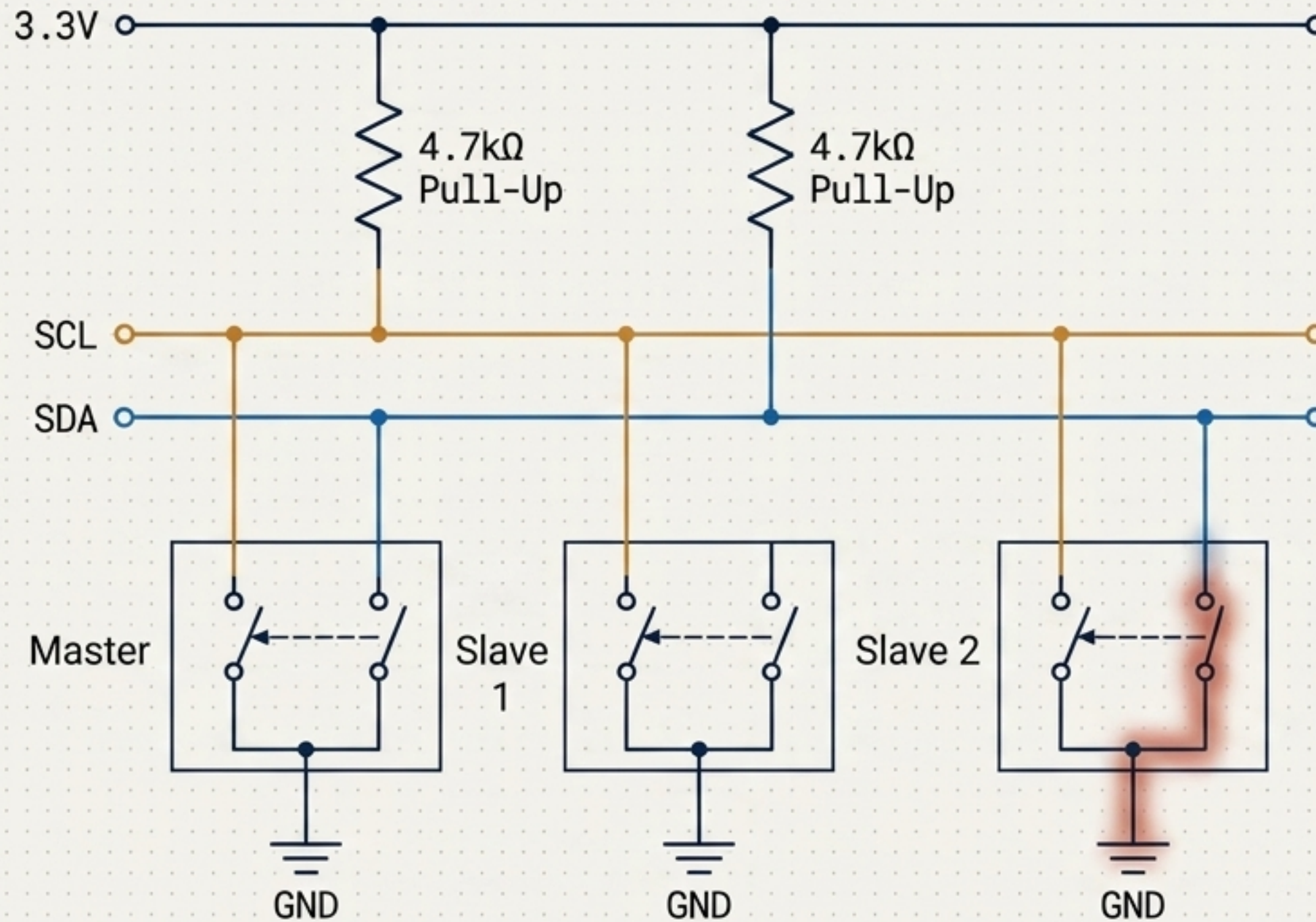
The 2-wire standard that has powered embedded communications for 40 years.

Developed by Philips in 1982, I2C (Inter-Integrated Circuit) solves a fundamental engineering constraint: connecting dozens of peripheral chips on a single board using the absolute minimum number of microcontroller pins.

Selecting the right protocol requires balancing speed against pin constraints.

I2C (The Multi-Tool)	SPI (The Sprinter)	UART (The Direct Line)
Pins: 2 Pins (+ GND) SDA (Serial Data)  SCL (Serial Clock) 	Pins: 4+ Pins (MOSI, MISO, CLK, CS) MOSI  MISO  CLK  CS 	Pins: 2 Pins (TX, RX) TX  RX 
Topology: Multi-Master / Multi-Slave 	Topology: 1-Master-to-N-Slaves 	Topology: Point-to-Point 
Max Devices: 127 1 2 7	Max Devices: Unlimited (Requires 1 CS pin per device) 	Max Devices: 1
Speed: Up to 3.4 Mbps 	Speed: 10+ Mbps 	Speed: ~1 Mbps 
Best for: Connecting many sensors on a pin-constrained bus.	Best for: High-speed data like SD cards or large displays.	Best for: Direct, simple chip-to-chip or PC-to-MCU serial connections.

Open-drain architecture prevents short circuits on a multi-master bus.



The Shared Bus Principle:

I2C is a half-duplex, shared system. Devices never push voltage into the line; they only pull it LOW to ground.

The Role of Pull-Ups:

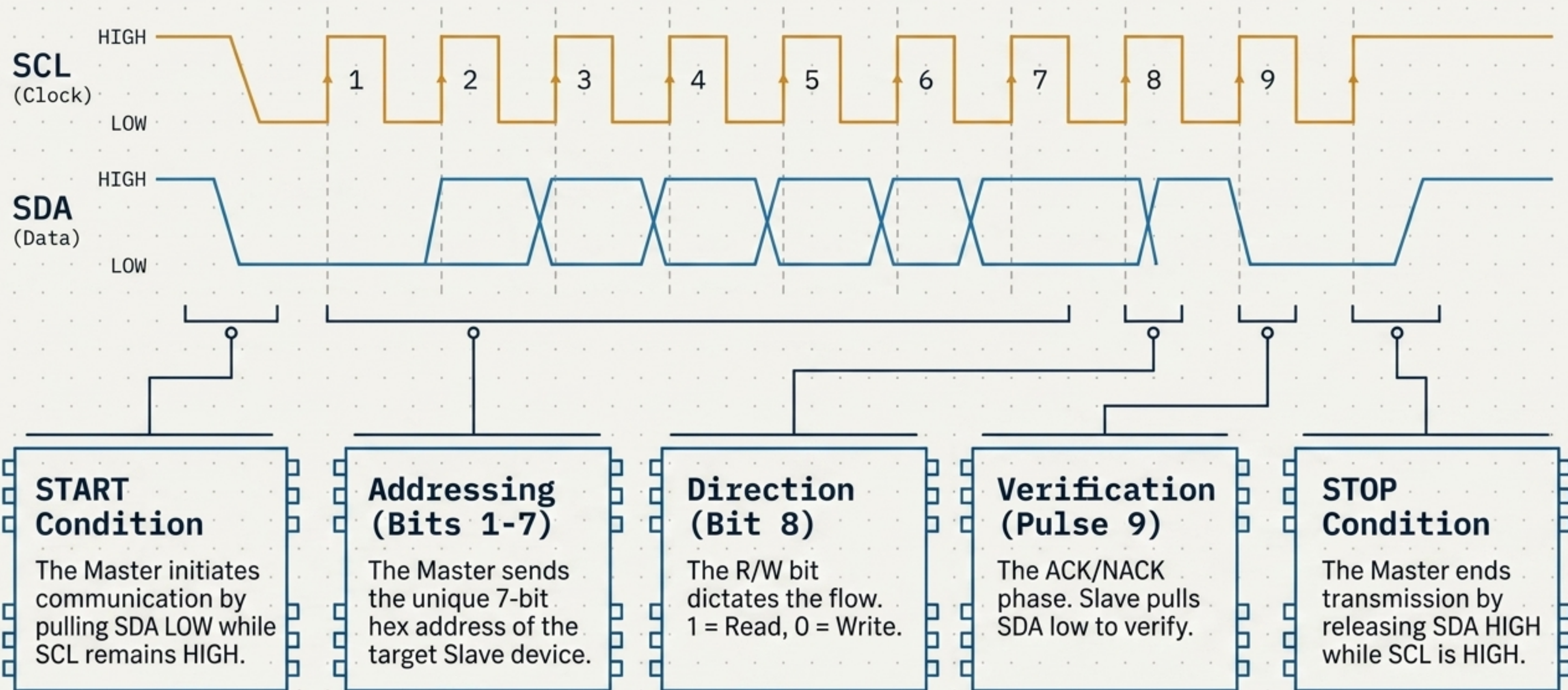
External 4.7kΩ resistors pull the lines HIGH to 3.3V when idle. If multiple masters try to talk at once, pulling to ground doesn't cause a short circuit.

Calculating Resistor Size

$$R_{pu} = \frac{V_{cc} - V_{ol}}{I_{ol}}$$

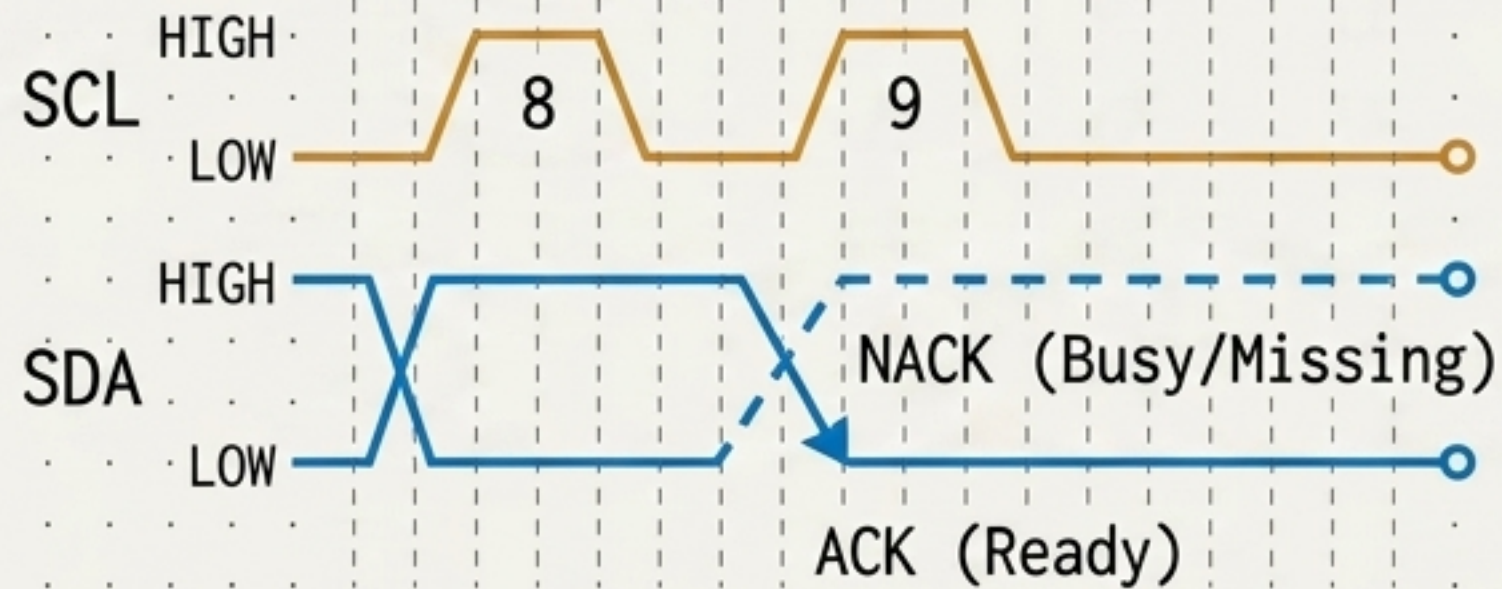
For a 3.3V system with an allowable $0.4V_{ol}$ and $3mA_{ol}$, the minimum safe resistor is $\sim 967\Omega$. 4.7kΩ is the safe, standard default for short buses.

Every transaction follows a strict chronological frame of pulses.



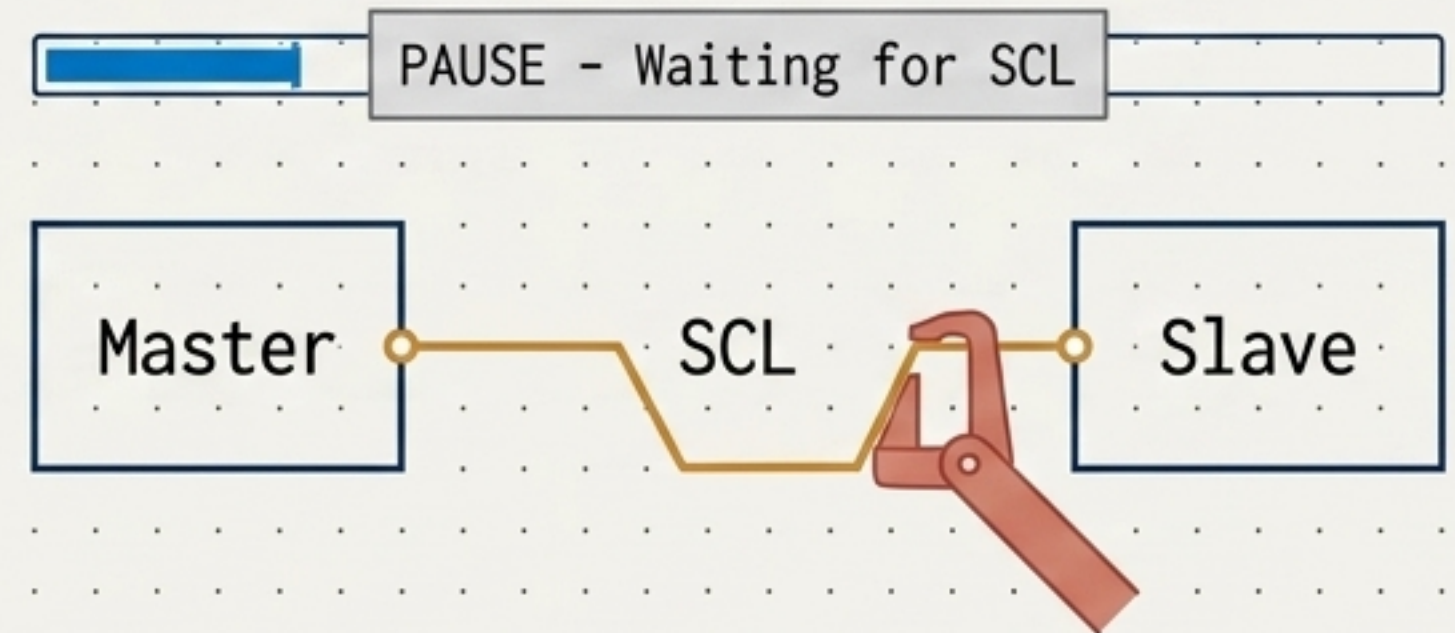
Handshaking and clock stretching keep mismatched speeds synchronized

ACK/NACK Mechanism



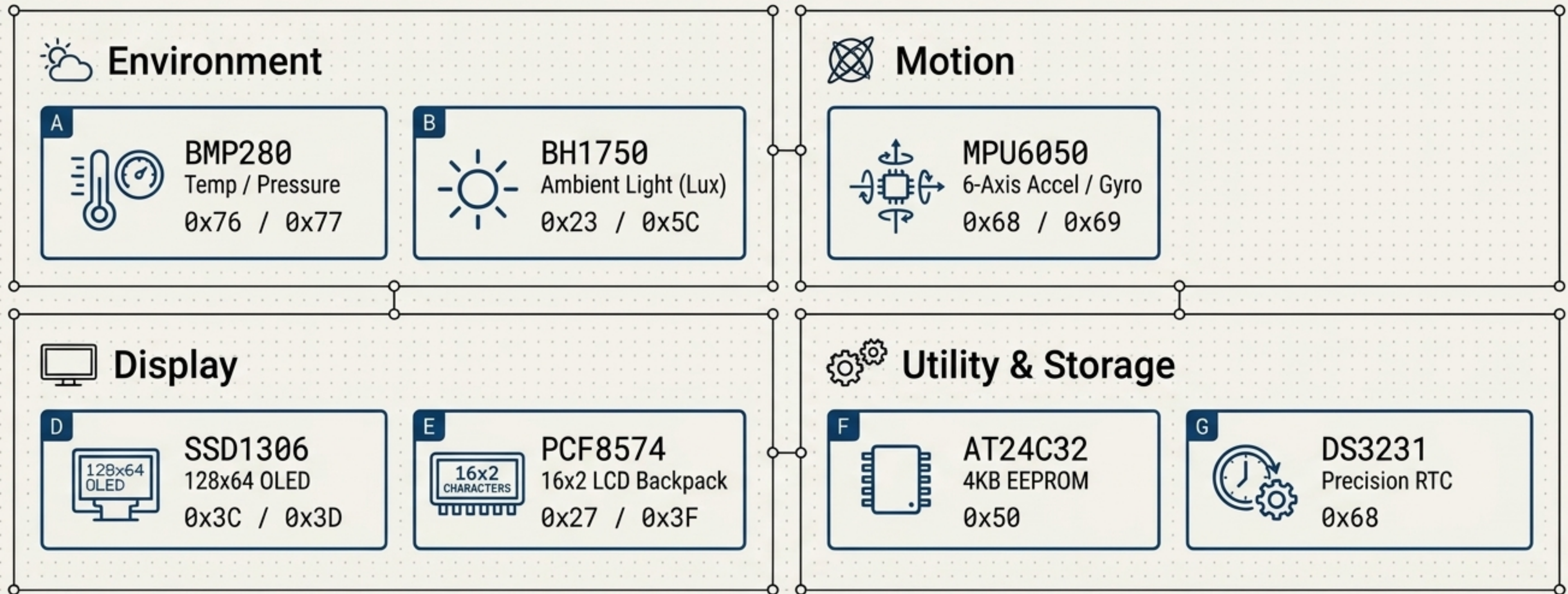
After 8 bits, the Master releases SDA. If the Slave is ready, it pulls SDA LOW on the 9th clock pulse (ACK). If SDA remains HIGH, it is a NACK, telling the Master the device is missing or busy.

Clock Stretching



A Slave can purposefully pull the SCL clock line LOW to tell the Master "Wait, I am processing data." The Master must wait for the Slave to release SCL before sending the next bit.

The sensor ecosystem is mapped by functional category and 7-bit addresses.



ADDRESS CONFLICT ALERT

The DS3231 and MPU6050 share the default 0x68 address. To prevent a bus collision, the MPU6050's AD0 pin must be pulled HIGH to shift its address to 0x69.

Software abstraction dictates the balance between control and speed.

STM32 (HAL Paradigm)
Strict & Granular Control

```
GPIO_MODE_AF_OD
```

```
HAL_I2C_Mem_Write(&hi2c1, dev_addr,  
reg_addr, size, data, timeout);
```

Requires manual Alternate Function Open-Drain setup. Relies on deep memory functions. Excellent for learning raw protocol mechanics and writing custom drivers from scratch.

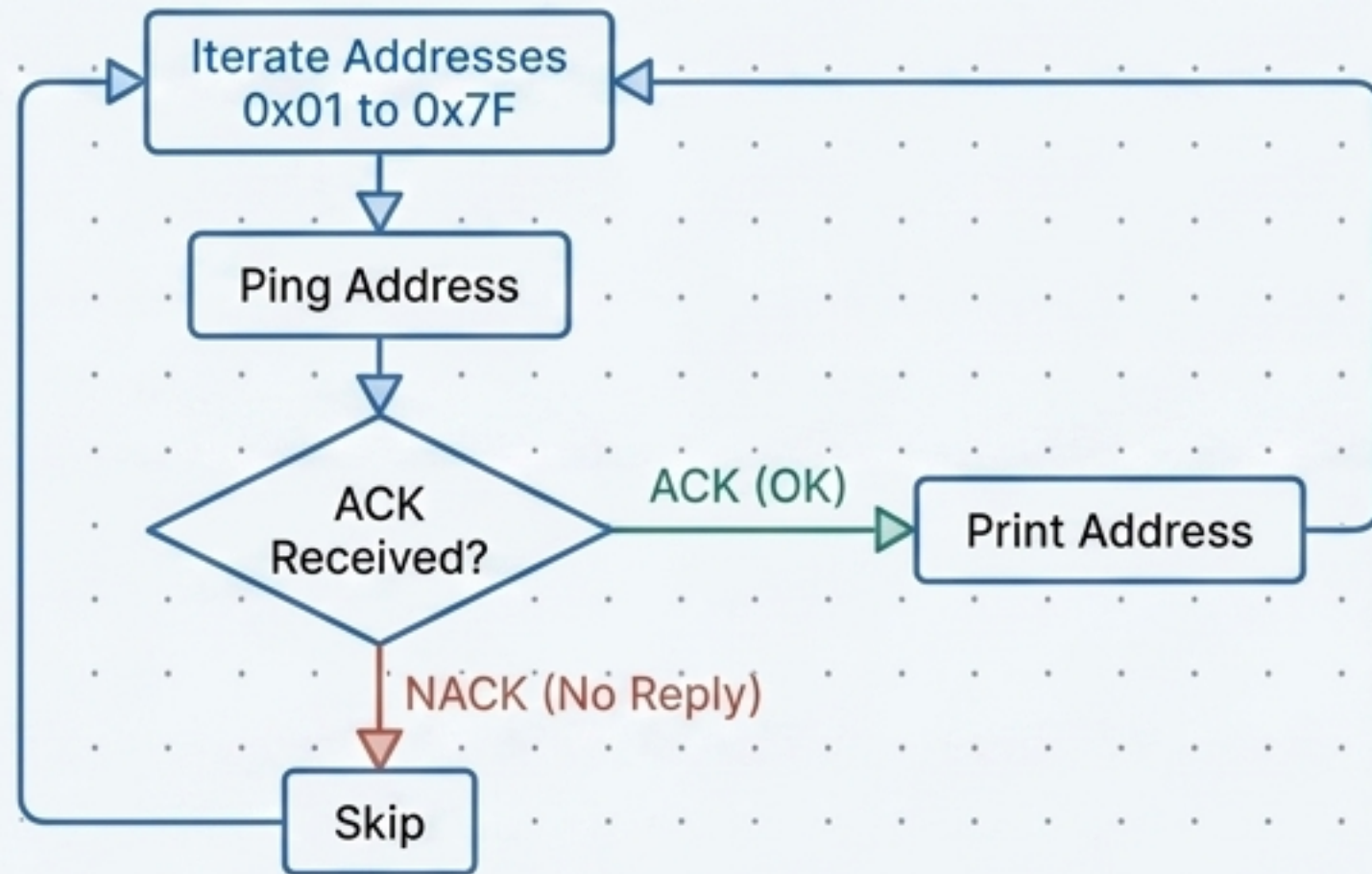
ESP32 (Arduino / Wire Paradigm)
Flexible & Abstracted

```
Wire.begin(21, 22, 400000);
```

```
Wire.beginTransmission(dev_addr);  
Wire.requestFrom(dev_addr, size);
```

Setup is executed in a single line with highly flexible GPIO mapping (e.g., SDA=21, SCL=22). Read/Write loops utilize abstracted methods for rapid prototyping and deployment.

The I2C Scanner is the ultimate diagnostic tool for bus conflicts and dead chips.



Scanning I2C bus...

Device found at 0x23
Device found at 0x3C
Device found at 0x76

Scan complete. 3 devices found.

The Check

If a device is physically connected, it will pull **SDA LOW (ACK)** on the 9th pulse. The MCU reads this as a 0 or **HAL_OK**.

The Utility

Run this before writing any driver code. It instantly identifies loose cables, missing pull-up resistors, burnt chips, or unshifted colliding addresses.

Pushing clock speeds requires tighter hardware tolerances.

Standard Mode



Standard Mode (100kHz)

Highly tolerant to noise and longer cables.
Multi-sensor loop takes ~30-50ms.
Ideal for lab environments and initial prototyping.
Default pull-up resistor: 4.7k Ω .

Fast Mode

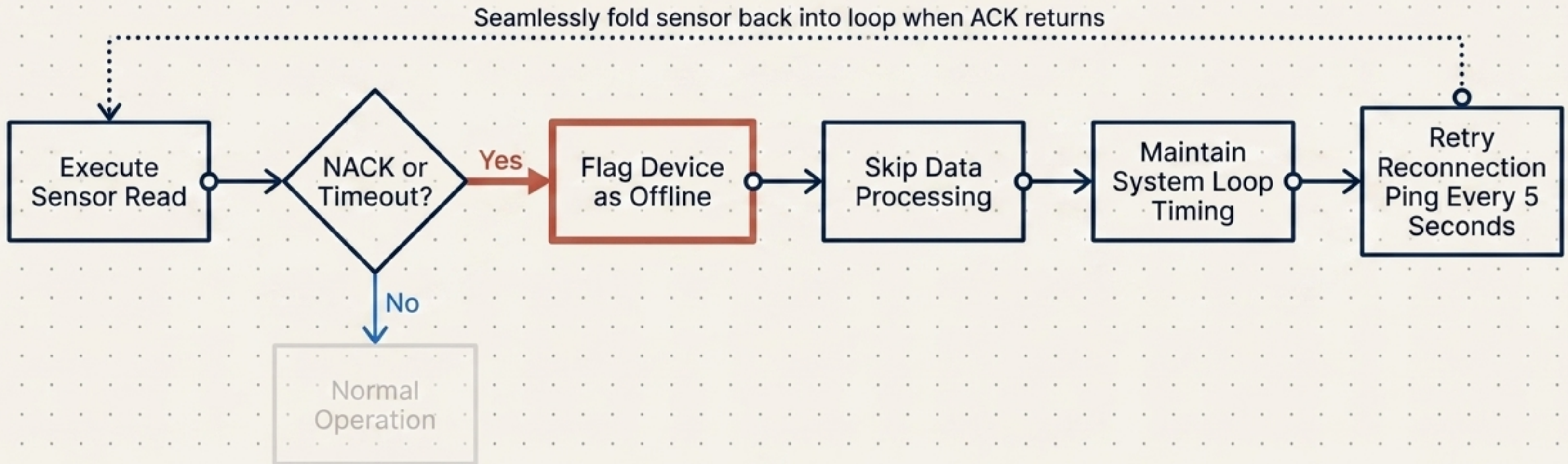


Fast Mode (400kHz)

Reduces the multi-sensor loop time to ~8-15ms (3-4x faster).
Faster rise time requirements mean you must lower the pull-up resistor value (e.g., to 2.2k Ω).
More susceptible to noise on long cables; requires tight hardware tolerances to prevent data corruption.

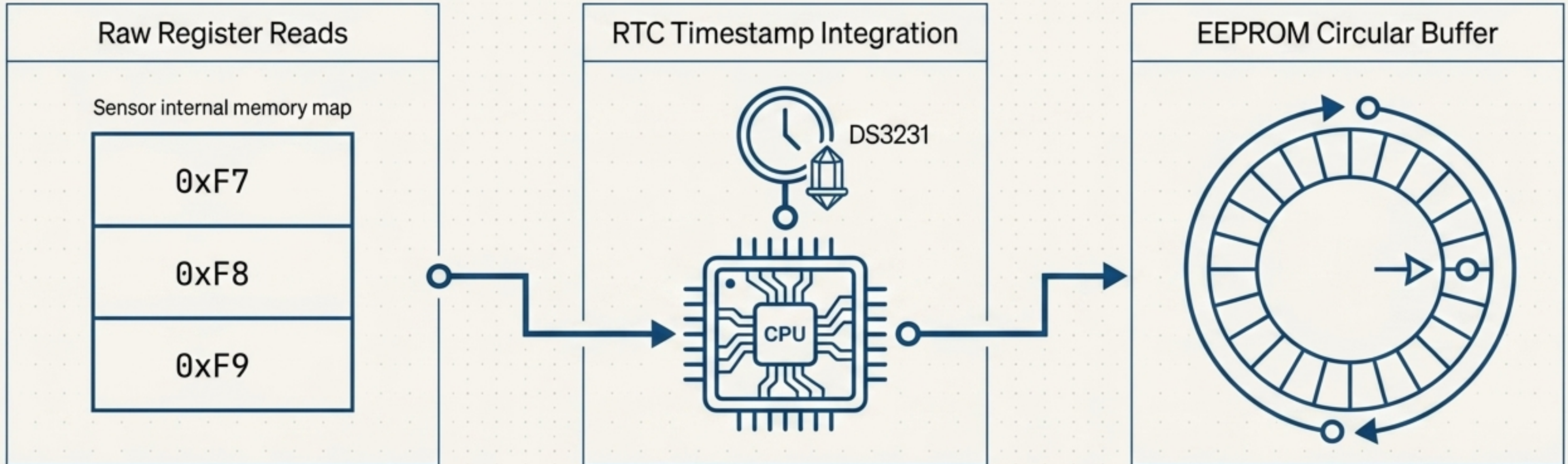
Robust systems flag hardware dropouts instead of crashing.

The Problem: Loose cables or power glitches cause standard library loops to hang infinitely waiting for an ACK that will never come.



Professional engineering prioritizes real-world troubleshooting.
Most failures are loose cables, not burnt chips.

Managing raw hex registers and non-volatile circular buffers



Bypassing libraries requires manual byte combination. A 24-bit reading requires reading three registers and bit-shifting:

$(\text{MSB} \ll 12) \mid (\text{LSB} \ll 4) \mid (\text{XLSB} \gg 4).$

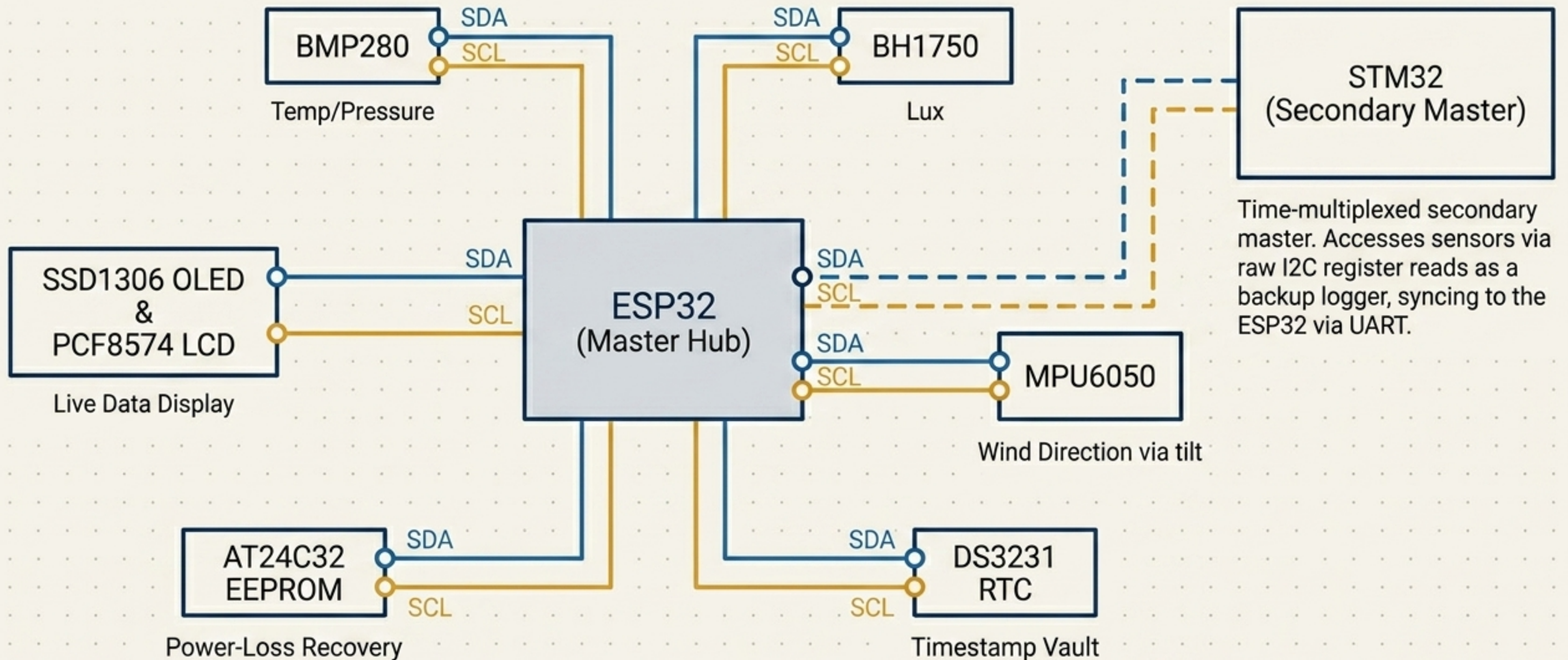
The DS3231 outputs time in BCD (Binary-Coded Decimal) format. The hex value 0x23 literally represents the number 23.

The AT24C32 has 32-byte page boundaries. Logging data requires wrapping pointers to create a circular buffer:

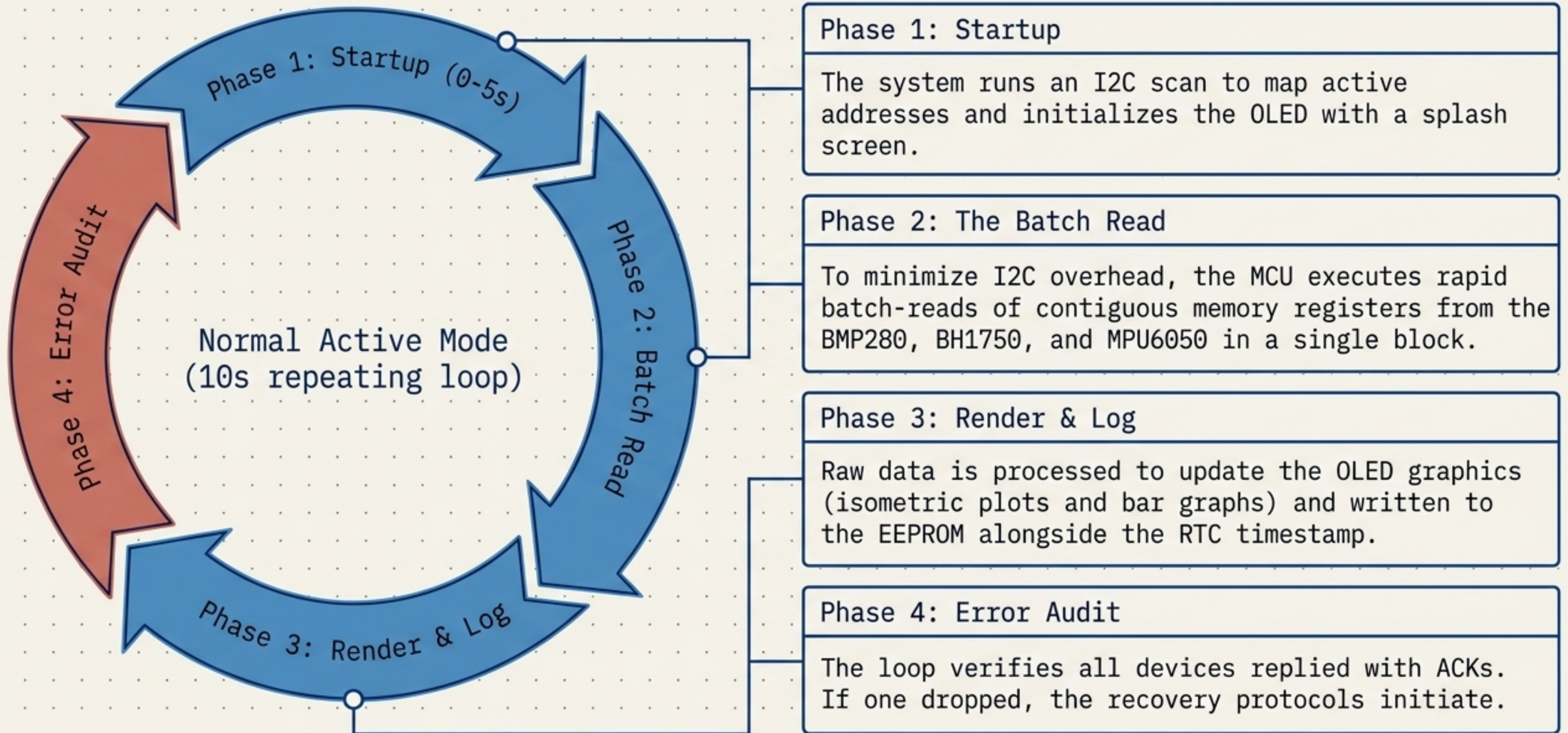
$\text{ptr} = (\text{ptr} + 1) \% 1000$

Newest weather data continuously overwrites the oldest without exceeding memory.

The Capstone Architecture: A multi-master, multi-sensor weather station.



Orchestrating the 10-second multi-sensor active loop

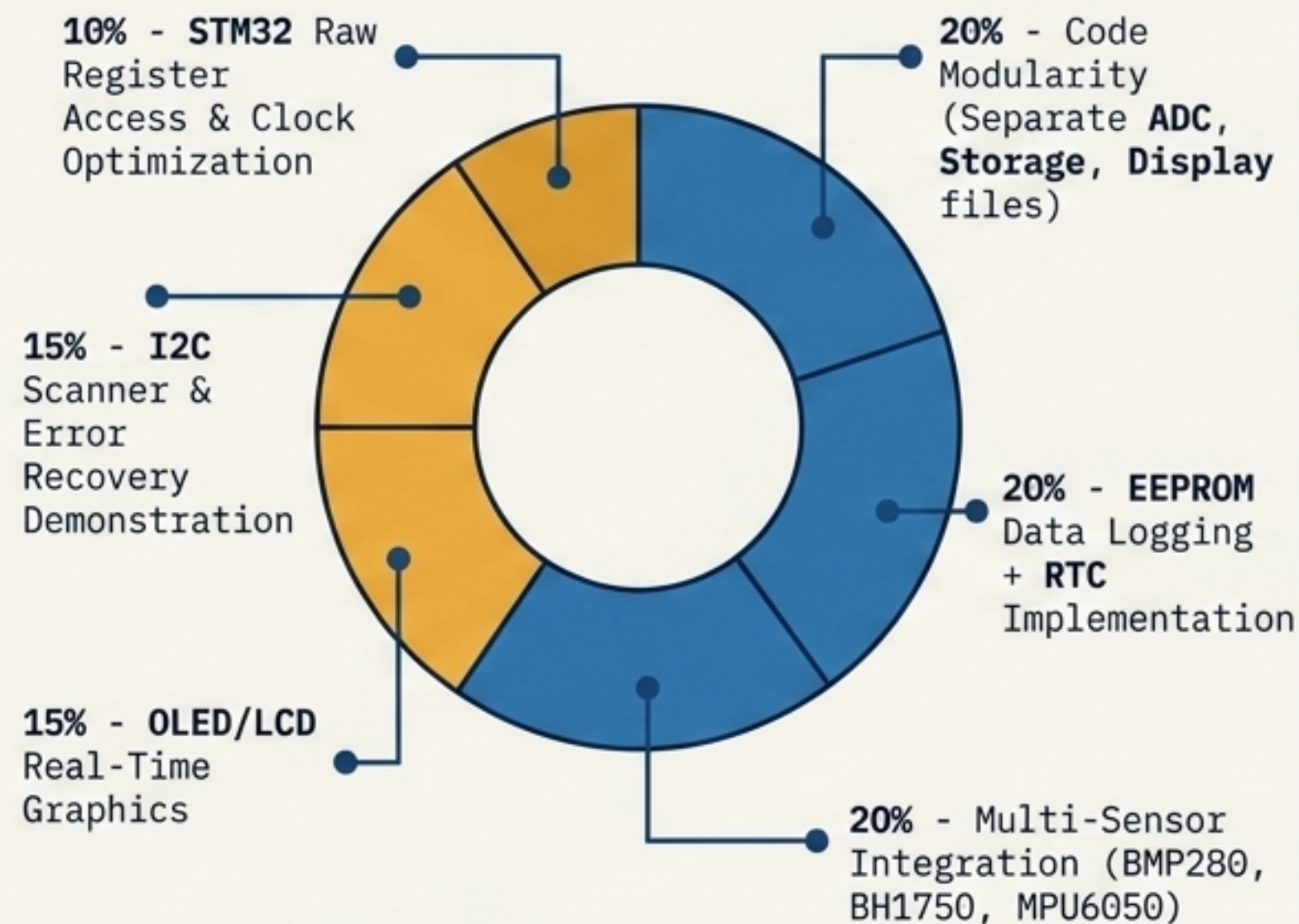


Project Deliverables and Assessment Rubric

The Video Deliverable

- ✓ **15-25** minute demonstration video.
- ✓ Uploaded via an unlisted YouTube link.
- ✓ Minimum **720p** resolution.
- ✓ Picture-in-Picture (**PiP**) required: display webcam, screen recording, and a live close-up of the physical hardware.

Rubric Breakdown



Production best practices to ensure a flawless hardware demonstration.

✓ DO (Best Practices)

Use a tripod or stable mount for the hardware close-up.

Ensure the screen recording code is legible (14pt font minimum).

Utilize an external microphone and ensure clear lighting.

Physically disconnect a sensor during the demo to prove the auto-recovery code works without crashing the system.

✗ DON'T (Penalties)

-20% Penalty: Failing to show your face (webcam required).

-20% Penalty: Missing physical hardware (No CGI or animations allowed).

-10% Penalty: Falling outside the required 15-25 minute duration.